

Pediatric vs Adult Dengue Severity Prediction with Subgroup-Specific SHAP Explanations

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Overview

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Background

Dengue severity models are usually trained on pooled, all-age cohorts - masking how children and adults actually differ.

Why

Pediatric and adult dengue patients show different physiology - plasma leakage, comorbidities, hemorrhagic patterns.

What is missing

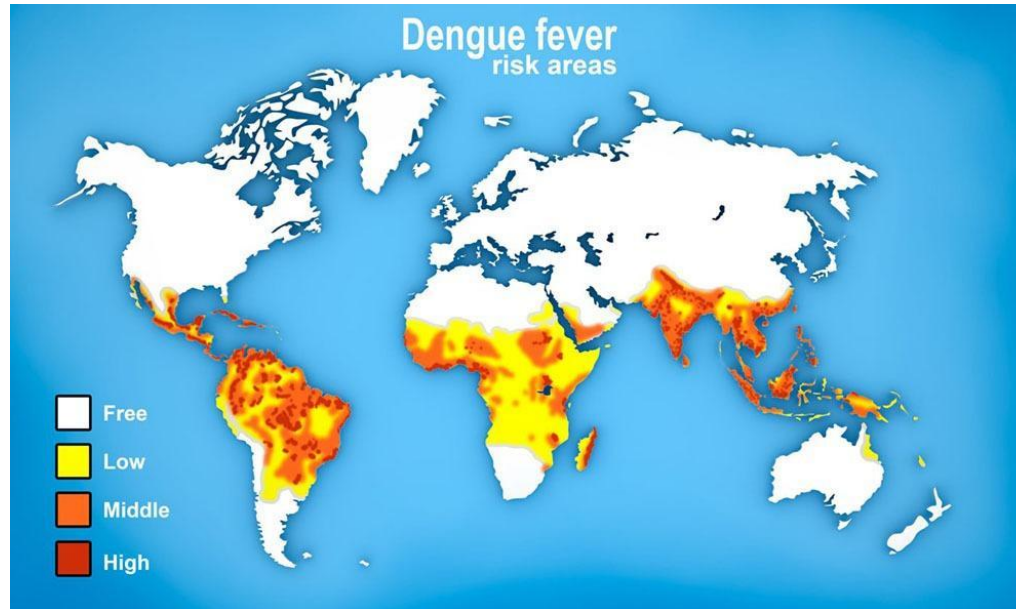
Most ML studies use a single pooled model and a single global SHAP summary. No dengue study quantitatively compares SHAP feature attribution between pediatric and adult subgroups.

What we have done

Age-stratified models + subgroup SHAP + an early-stage risk-assessment dashboard.



Introduction



Why this matters ?

Dengue is a mosquito-borne viral infection (*Aedes aegypti*, *Aedes albopictus*) and a major public health burden across tropical and subtropical regions.

Early identification of patients likely to progress to severe disease is critical for timely fluid management and supportive care.



Introduction

Pooled-model assumption : Most prior ML work treats age as just one input feature, not a stratification variable.

Clinical reality : Children show higher rapid plasma leakage; adults show more comorbidities and hemorrhagic signs.

Our thesis position : Compare pooled vs. age-stratified models and their SHAP explanations directly, then expose the result via a dashboard.



Problem Statement

Age-pooled model training: Classifiers are typically trained on pooled pediatric + adult data, assuming one feature–severity relationship fits all ages.

Global feature attribution: SHAP explanations are computed for the whole pooled cohort, averaging out subgroup-level differences.

Misalignment with clinical practice: Clinicians judge children and adults against age-specific norms; pooled models give one generic boundary.

Lack of explainable deployment: Trained models are rarely wired into a system that supports patient-level explanation or structured data collection.

Research question

Do subgroup-trained models and their SHAP explanations differ meaningfully from an age-pooled baseline for dengue severity prediction -and can that difference be made usable through a dashboard?



Objectives

1. Develop pediatric, adult, and pooled ML models for severe-dengue prediction using classifiers.
2. Generate subgroup-specific SHAP explanations, identify model-derived feature splits
3. An explainable web dashboard for longitudinal data collection.



Related Works

Ref	Focus	Cohort	Methods	Outcome Definition	Gaps
[1]	Early-phase dengue diagnosis	Decision-tree algorithms	Diagnosis + outcome(early illness)	Diagnosis + outcome(early illness)	Age-aware but pre-SHAP; no subgroup specific explain-ability
[2]	Pediatric severity using early labs	Thailand, pediatric cohort	Logistic ML (misclassification cost ratio 1:10)	Dengue severity from early labs	Pediatric only; no adult comparison subgroup
[3]	Clinical pattern discovery	Brazil, n=527	K-means+hierarchical clustering	Clinical phenotypes (not WHO severity)	Age identified as top discriminator; no subgroup specific prediction

Related Works

Ref	Focus	Cohort	Methods	Outcome Definition	Gaps
[4]	Demographics + lab ML	Taiwan, n=798 (138 severe	ANN, RF,GBM,SVM,LR +SHAP	Severe dengue (binary	ANN AUC ~ 0.83; pooled cohort, global SHAP only
[5]	Stacking ensemble + SHAP	Mixed, 17 features	11 ML models + Staking + SMOTE	Severe dengue	77% baseline / 83% post-SMOTE accuracy; global SHAP only
[6]	Stacked ensemble + dual XAI	Bangladesh, n=1523	Stacked ensemble + SMOTEENN + SHAP/LIME	Dengue severity (multiclass)	96.38% accuracy; global explanation only

Related Works

Ref	Focus	Cohort	Methods	Outcome Definition	Gaps
[7]	SHAP-based factor analysis	Vietnam+Bangladesh	XGBoost+SHAP plots	WHO severity classes	Global SHAP only; no age-conditioned explanation
[8]	Hematology driven severity	Public dataset n=1450	LR,RF,SVM, CGBoost +SHAP	WHO severity classes	XGBoost macro-F1 = 0.59; global SHAP
[9]	Optimized feature diagnosis	Mixed, RFE selected features	RF + RFE + SHAP + LIME	Dengue diagnosis/severity	RF 99.67% accuracy; global XAI

Proposed Methodology

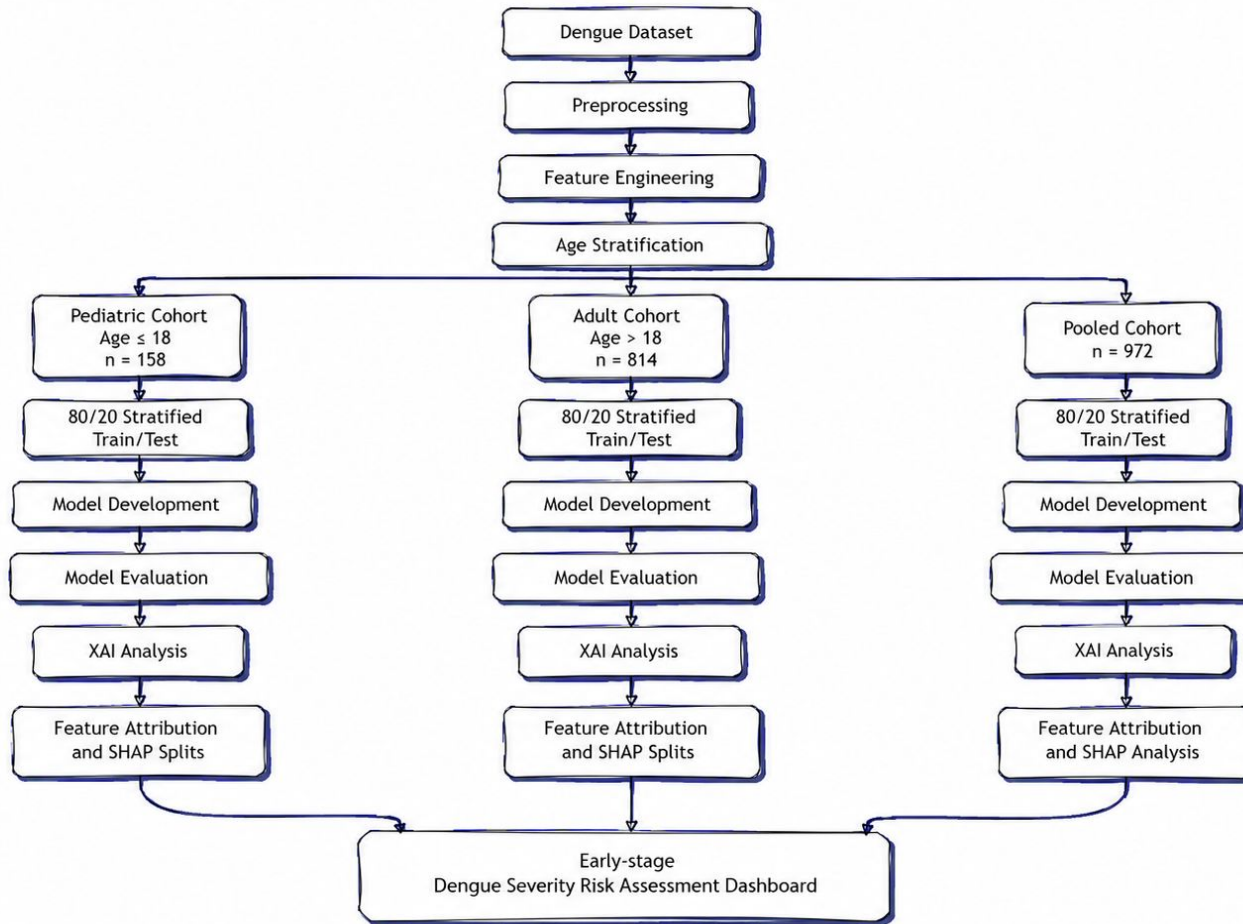


Figure: Workflow overview



Methodology: Data Preparation

- Dataset Collection: Comprehensive Dengue Hematology and Clinical Dataset from Bangladesh for Machine Learning and Diagnostic Research [10]
- 9 □ Core clinical/demographic predictors (WBC, HCT, RBC, Lymph %, Neut %, AST, ALT, Age, Gender)
- 972 □ Total NS1-positive records (from 1,329 raw records)
- 158 / 814 □ Pediatric ($\leq 18y$) / Adult ($> 18y$) split
- 80:20 □ Stratified train/test split, random_state = 42

Feature	Category	Description	Role in Model
Age	Demographic	Patient age	Predictor + cohort split
Gender	Demographic	Patient gender	Predictor
WBC	Hematological	White blood cell count	Predictor
HCT	Hematological	Hematocrit	Predictor
RBC	Hematological	Red blood cell count	Predictor
Lymphocyte %	Differential	Lymphocyte proportion	Predictor
Neutrophil %	Differential	Neutrophil proportion	Predictor
AST	Biochemical	Aspartate amino-transferase	Predictor
ALT	Biochemical	Alanine amino-transferase	Predictor
PLT	Hematological	Platelet count	Target construction; excluded



Methodology: Model & Evaluation

- 5 classifiers per cohort: Logistic Regression, Random Forest, XGBoost, LightGBM, Stacking Ensemble (LR meta-learner).
- Metrics: Accuracy, AUC-ROC, Sensitivity, Specificity, F1, Matthews Correlation Coefficient (MCC).
- Explainability: SHAP TreeExplainer on best performed model; depth-1 decision trees fitted to SHAP>0 values to extract feature splits; Kendall's τ -b for pediatric–adult rank comparison.



Methodology: Dashboard development

- Programming Languages
 - Javascript
 - Python
- Stack used
 - Frontend: ReactJs
 - Backend: FastApi + Cohort specific best performed models
 - Database: PostgreSQL



Result - Model Performance Summary

Cohort	Best Model	AUC-ROC	Accuracy	Sensitivity	Specificity	F1-Score	MCC
Pediatric (≤ 18)	Random Forest	0.961	90.6%	66.7%	100.0%	0.800	0.768
Pediatric (≤ 18)	Stacking Classifier	0.947	90.6%	66.7%	100.0%	0.800	0.768
Pediatric (≤ 18)	XGBoost	0.899	84.4%	77.8%	87.0%	0.737	0.628
Adult (> 18)	Random Forest	0.892	84.7%	78.1%	90.0%	0.820	0.690
Adult (> 18)	Stacking Classifier	0.882	83.4%	79.5%	86.7%	0.811	0.664
Adult (> 18)	XGBoost	0.856	78.5%	79.5%	77.8%	0.768	0.570

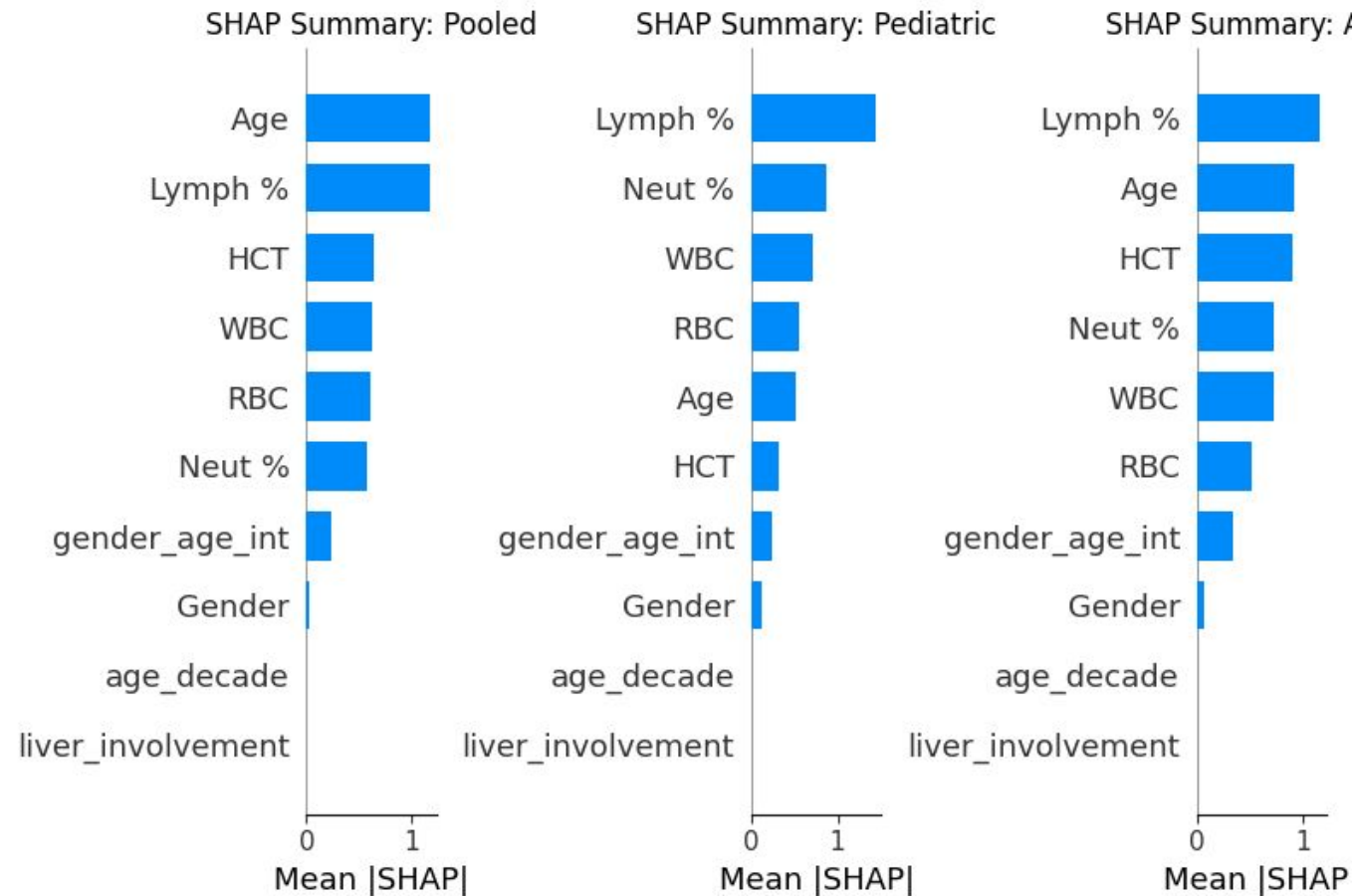


Result - Cohort Specific Selected Model

Cohort	Best Model	Accuracy	AUC	F1	MCC
Pediatric	Stacking / Random Forest	90.6%	0.947 / 0.961	0.800	0.768
Adult	Random Forest	84.7%	0.892	0.820	0.690
Pooled	Random Forest	79.0%	0.860	0.739	0.566



Result - SHAP Feature Attribution



Result - Model-Derived SHAP Feature Splits

TWO-COLUMN COMPARISON

PEDIATRIC SPLITS (SHAP > 0)

- Lymphocyte % - split at > 44.50%
- Neutrophil % - split at $\leq 48.00\%$
- WBC - split at $\leq 4.34 \times 10^3/\mu\text{L}$

ADULT SPLITS (SHAP > 0)

- Lymphocyte % - split at > 24.99%
- HCT - split at > 40.65%
- Age - split at > 31.00 years



Result - Web Application

Early-stage Explainable Dengue Risk Assessment Dashboard

Research prototype for structured longitudinal data collection

Disclaimer: The dashboard is an early-stage research interface intended for model demonstration and structured data collection. It is not intended for clinical diagnosis, treatment selection, or patient management. Its predictions are based on a retrospective dataset and require independent and prospective validation.

Patient ID

p001

Gender

Male (0)

Age (years)

24

Pediatric cohort: ≤ 18 · Adult cohort: > 18

WBC ($\times 10^3/\mu\text{L}$)

5.15

Normal: 4–10 · Leukopenia flag: < 4.0

Lymphocyte (%)

19.6

Lymphopenia flag: < 20%

Neutrophil (%)

72.2

Neutrophilia flag: > 70%

Hematocrit (HCT) (%)

35

Normal: 36–50%

RBC ($\times 10^6/\mu\text{L}$)

3.6

Normal: 3.8–6.1

AST (U/L)

46

Liver flag: > 40 U/L · Dataset median: 48

ALT (U/L)

36

Liver flag: > 40 U/L · Dataset median: 35

Platelets (PLT) ($\times 10^3/\mu\text{L}$)

47

Severe: < 50 · Moderate: 50–100

Risk Assessment

Adult Model Active

8%

Low-risk pattern

Generate PDF Report

Load History

+ Run & Store Assessment Snapshot

Longitudinal Patient Timeline

The dashboard provides longitudinal visualization of repeated laboratory measurements and independently generated model predictions. Temporal prediction is outside the scope of the current model.

Assessment 1 (2026-08-14)

9%

Low-risk pattern

Assessment 2 (2026-08-14)

8%

Low-risk pattern

Assessment 3 (2026-08-14)

8%

Low-risk pattern

Change Analysis (Current vs Previous)

Model prediction changed from 8% to 8%.

Feature	Previous (Assessment 2 (2026-08-14))	Current (Assessment 3 (2026-08-14))	Change
Age	24	24	0.0%
WBC	5.15	5.15	0.0%
HCT	35	35	0.0%
RBC	3.6	3.6	0.0%
Lymphocyte %	19.6	19.6	0.0%
Neutrophil %	72.2	72.2	0.0%
ALT	36	36	0.0%
AST	46	46	0.0%

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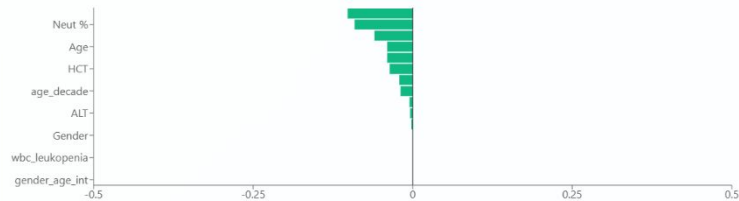
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Result-Web Application

Why this prediction? (SHAP Explanation)

This panel displays model-derived SHAP splits as contextual research findings. It explains how each patient feature contributed to the model's prediction. Features pushing the risk higher are shown in red (to the right), and features reducing risk are shown in green (to the left). Note: probability comes strictly from the trained model and is never calculated from the SHAP split.



Increasing Risk

None

Reducing Risk

Lymph % (19.6)	-0.102
Neut % (72.2)	-0.091
lymph_low (1)	-0.06
Age (24)	-0.04
neut_high (1)	-0.04
HCT (35)	-0.036
WBC (5.15)	-0.021
age_decade (2)	-0.019
ast_alt_ratio (1.28)	-0.005
ALT (36)	-0.004
RBC (3.6)	-0.002
Gender (0)	0
AST (46)	0
wbc_leukopenia (0)	0
liver_involvement (1)	0
gender_age_int (0)	0

Pediatric vs Adult Comparison Mode

This section demonstrates why age-specific models are necessary. The same patient characteristics receive different model contributions and risk predictions depending on the age-specific model used.

Metric	Pediatric Model	Adult Model
Algorithm	XGBoost	XGBoost
Predicted Risk	17%	8%
Top Feature	lymph_low	Lymph %
SHAP Contribution	-0.113	-0.102
Relevant Split	Model Split	Model Split



Conclusion & Future Work

Age-stratified models outperformed the pooled baseline (pediatric: 90.6% accuracy / AUC 0.947 Stacking; adult: 84.7% accuracy / AUC 0.892 Random Forest). SHAP rankings were highly correlated across age groups ($\tau_b = 0.900$), yet model-derived feature-split thresholds differed meaningfully by age - confirming that pooled models can mask clinically relevant, age-specific decision behavior.

FUTURE WORK

- External, prospective validation on independent, geographically distinct datasets
- Larger pediatric cohort to strengthen subgroup robustness
- WHO-adjudicated severity labels in place of the PLT-derived proxy
- Dashboard-driven longitudinal data collection enabling future temporal/sequential modeling (e.g., RNNs)



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